

Multivariate Analysis - PCA and EFA

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Table of contents

1. Introduction	1
1.1. Project Libraries and Installation	2
1.2. Initial description of the data (a)	2
2. Principal Component Analysis (b)	4
2.1. Variable selection	4
2.2. Suitability of the data	4
2.3. Running the PCA	5
2.4. Number of components (c)	5
Kaiser criterion	5
60% of cumulative variance	6
Horn's parallel analysis	7
Decision	8
2.5. Interpreting the components	8
2.6. Graphical representation	11
3. Selecting for the Exploratory Factor Analysis (d)	12
3.1. Number of factors	12
3.2. Which variables to keep	13
4. Exploratory Factor Analysis (e)	13
4.1. Estimation	13
4.2. Model fit	14
4.3. Interpreting the factors	15

1. Introduction

Analyse the variables below from the dataset 'Holzinger-Swineford 1939' with multivariate methods such as Principal Component Analysis (PCA for short) and Exploratory Factor Analysis (EFA).

1. Visual (x1)
2. Cubos (x2)
3. Índice de Lozenges (x3)
4. Compresión de párrafos (x4)
5. Completar oraciones (x5)

6. Significado de palabras (x6)
7. Velocidad para sumar (x7)
8. Velocidad para contar puntos (x8)
9. Discriminación acelerada de mayúsculas rectas y curvas (x9)

1.1. Project Libraries and Installation

```
libs <- c("lavaan", "psych", "GPArotation", "FactoMineR",
  "factoextra", "corrplot", "skimr", "tidyverse")

to_install <- setdiff(libs, rownames(installed.packages()))
if (length(to_install)) install.packages(to_install)

invisible(lapply(libs, library, character.only = TRUE))
```

The dataset comes with lavaan.

```
library(lavaan)

data <- HolzingerSwineford1939
```

1.2. Initial description of the data (a)

We first glimpse at the data with `glimpse(data)` and `skim(data)`

```
glimpse(data)
```

```
Rows: 301
Columns: 15
$ id      <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 18, 19, ...
$ sex     <int> 1, 2, 2, 1, 2, 2, 1, 2, 2, 2, 1, 1, 2, 2, 1, 2, 2, 1, 2, ...
$ ageyr   <int> 13, 13, 13, 13, 12, 14, 12, 12, 13, 12, 12, 12, 12, 12, 12, 12, ...
$ ageemo  <int> 1, 7, 1, 2, 2, 1, 1, 2, 0, 5, 2, 11, 7, 8, 6, 1, 11, 5, 8, 3, 1...
$ school  <fct> Pasteur, Pasteur, Pasteur, Pasteur, Pasteur, Pasteur, Pasteur, ...
$ grade   <int> 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, ...
$ x1      <dbl> 3.333333, 5.333333, 4.500000, 5.333333, 4.833333, 5.333333, 2.8...
$ x2      <dbl> 7.75, 5.25, 5.25, 7.75, 4.75, 5.00, 6.00, 6.25, 5.75, 5.25, 5.7...
$ x3      <dbl> 0.375, 2.125, 1.875, 3.000, 0.875, 2.250, 1.000, 1.875, 1.500, ...
$ x4      <dbl> 2.333333, 1.666667, 1.000000, 2.666667, 2.666667, 1.000000, 3.3...
$ x5      <dbl> 5.75, 3.00, 1.75, 4.50, 4.00, 3.00, 6.00, 4.25, 5.75, 5.00, 3.5...
$ x6      <dbl> 1.2857143, 1.2857143, 0.4285714, 2.4285714, 2.5714286, 0.857142...
$ x7      <dbl> 3.391304, 3.782609, 3.260870, 3.000000, 3.695652, 4.347826, 4.6...
$ x8      <dbl> 5.75, 6.25, 3.90, 5.30, 6.30, 6.65, 6.20, 5.15, 4.65, 4.55, 5.7...
$ x9      <dbl> 6.361111, 7.916667, 4.416667, 4.861111, 5.916667, 7.500000, 4.8...
```

```
skim(data)
```

```
Name                data
Number of rows      301
Number of columns    15
```

Column type frequency:

```
factor              1
numeric             14
```

```
Group variables      None
```

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
school	0	1	FALSE	2	Pas: 156, Gra: 145

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
id	0	1	176.55	105.94	1.00	82.00	163.00	272.00	351.00	■■■■■■■■■■
sex	0	1	1.51	0.50	1.00	1.00	2.00	2.00	2.00	■_■■■■■■■
ageyr	0	1	13.00	1.05	11.00	12.00	13.00	14.00	16.00	■■■■■■■■■■
agemo	0	1	5.38	3.45	0.00	2.00	5.00	8.00	11.00	■■■■■■■■■■
grade	1	1	7.48	0.50	7.00	7.00	7.00	8.00	8.00	■_■■■■■■■
x1	0	1	4.94	1.17	0.67	4.17	5.00	5.67	8.50	_■■■■■■■■
x2	0	1	6.09	1.18	2.25	5.25	6.00	6.75	9.25	_■■■■■■■■
x3	0	1	2.25	1.13	0.25	1.38	2.12	3.12	4.50	■■■■■■■■■■
x4	0	1	3.06	1.16	0.00	2.33	3.00	3.67	6.33	_■■■■■■■■
x5	0	1	4.34	1.29	1.00	3.50	4.50	5.25	7.00	_■■■■■■■■
x6	0	1	2.19	1.10	0.14	1.43	2.00	2.71	6.14	■■■■■■■■■■
x7	0	1	4.19	1.09	1.30	3.48	4.09	4.91	7.43	_■■■■■■■■
x8	0	1	5.53	1.01	3.05	4.85	5.50	6.10	10.00	■■■■■■■■■■
x9	0	1	5.37	1.01	2.78	4.75	5.42	6.08	9.25	■■■■■■■■■■

The correlation matrix shows how strongly x1 to x9 are related.

```
df <- data |> select(x1:x9)
corrplot(cor(df), method = "number")
```



2. Principal Component Analysis (b)

2.1. Variable selection

We use the nine test scores x1 to x9: continuous, complete, and measuring the abilities of interest. The other columns are identifiers or demographics, so we exclude them.

2.2. Suitability of the data

We check that the variables are correlated enough for PCA:

- **KMO:** acceptable above 0.6.
- **Bartlett's test:** a significant p-value means the variables are correlated.

```
KMO(df)
```

```
Kaiser-Meyer-Olkin factor adequacy
Call: KMO(r = df)
Overall MSA = 0.75
MSA for each item =
  x1  x2  x3  x4  x5  x6  x7  x8  x9
0.81 0.78 0.73 0.76 0.74 0.81 0.59 0.68 0.79
```

```
cortest.bartlett(cor(df), n = nrow(df))
```

```
$chisq
[1] 904.0971

$p.value
[1] 1.912079e-166

$df
[1] 36
```

KMO = 0.75 and Bartlett's test is significant ($\chi^2 = 904.1$, $df = 36$, $p < 0.001$), so PCA is appropriate.

2.3. Running the PCA

Variables are standardised (`scale.unit = TRUE`) so each test weighs equally. We keep all 9 components (`ncp`) so the scree plots show the full spectrum.

```
pca <- PCA(df, scale.unit = TRUE, ncp = ncol(df), graph = FALSE)
```

2.4. Number of components (c)

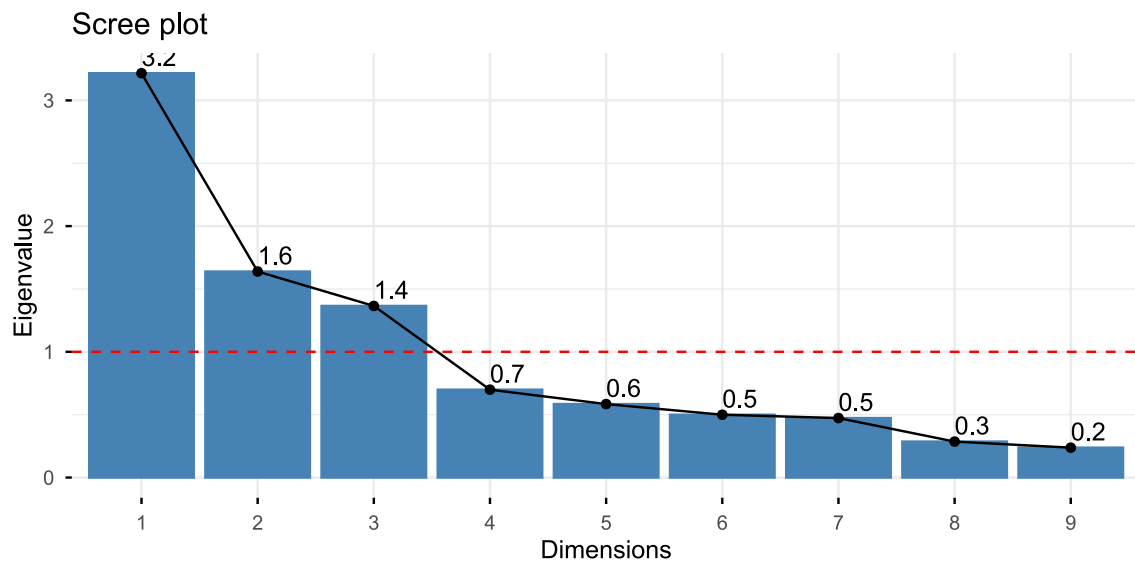
```
get_eigenvalue(pca)
```

	eigenvalue	variance.percent	cumulative.variance.percent
Dim.1	3.2163442	35.737158	35.73716
Dim.2	1.6387132	18.207925	53.94508
Dim.3	1.3651593	15.168437	69.11352
Dim.4	0.6989185	7.765761	76.87928
Dim.5	0.5843475	6.492750	83.37203
Dim.6	0.4996872	5.552080	88.92411
Dim.7	0.4731021	5.256690	94.18080
Dim.8	0.2860024	3.177804	97.35860
Dim.9	0.2377257	2.641396	100.00000

Kaiser criterion

Keep components with eigenvalue above 1, i.e. those explaining more than a single standardised variable.

```
fviz_eig(pca, addlabels = TRUE, choice = "eigenvalue") +  
  geom_hline(yintercept = 1, linetype = "dashed", colour = "red")
```



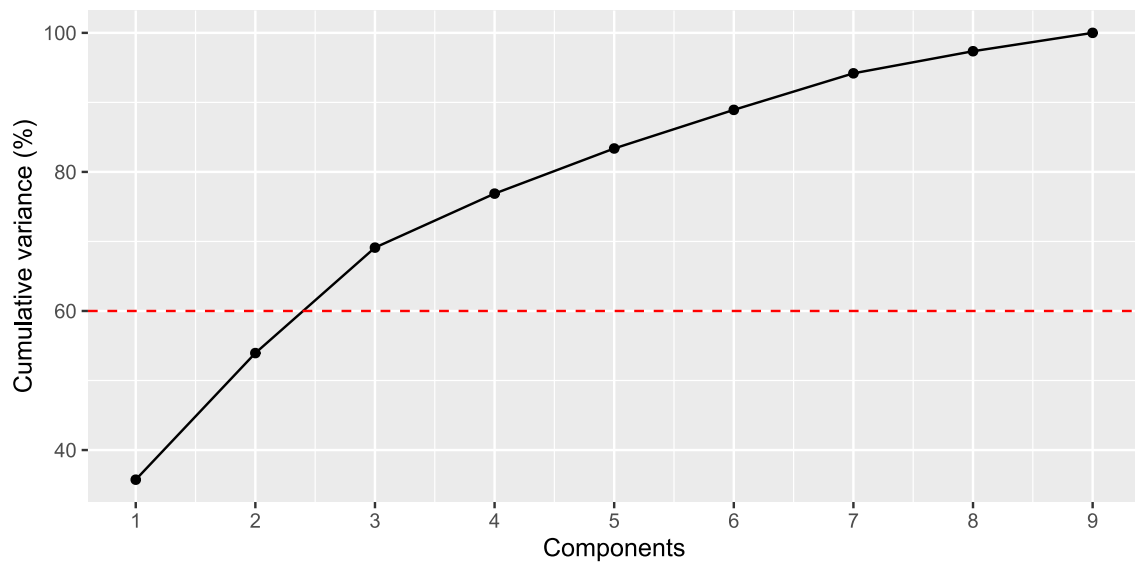
Three eigenvalues exceed 1 (3.22, 1.64, 1.37). Kaiser suggests **3 components**.

60% of cumulative variance

Keep the fewest components that together explain at least 60% of the variance.

```
eig <- get_eigenvalue(pca)

ggplot(eig, aes(x = seq_len(nrow(eig)), y = cumulative.variance.percent)) +
  geom_line() +
  geom_point() +
  geom_hline(yintercept = 60, linetype = "dashed", colour = "red") +
  scale_x_continuous(breaks = seq_len(nrow(eig))) +
  labs(x = "Components", y = "Cumulative variance (%)")
```

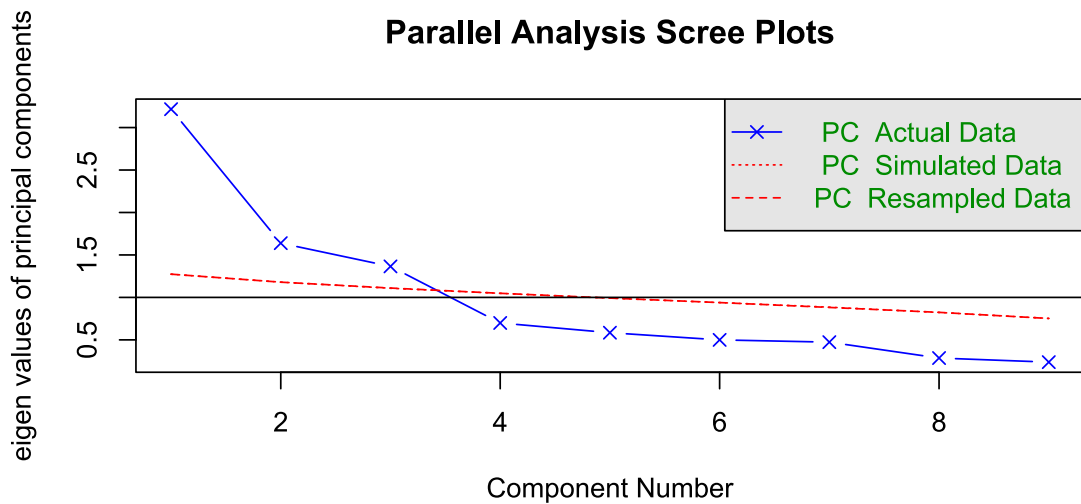


Two components explain 53.9%, three explain 69.1%. The 60% rule suggests **3 components**.

Horn's parallel analysis

Compare each observed eigenvalue with the eigenvalues of random data of the same size. Keep components whose eigenvalue is larger than the random one.

```
set.seed(123)
fa.parallel(df, fa = "pc", n.iter = 500)
```



```
Parallel analysis suggests that the number of factors = NA and the number of components = 3
```

Observed eigenvalues 3.22, 1.64 and 1.37 beat the random ones (1.27, 1.18, 1.11), but the fourth (0.70) does not (1.05). Parallel analysis suggests **3 components**.

Decision

Criterion	Components
Kaiser	3
60% cumulative variance	3
Horn's parallel analysis	3

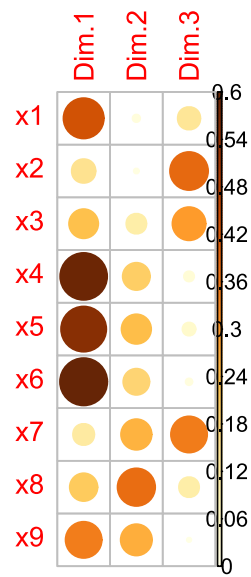
All three criteria agree, so we keep **three components**.

2.5. Interpreting the components

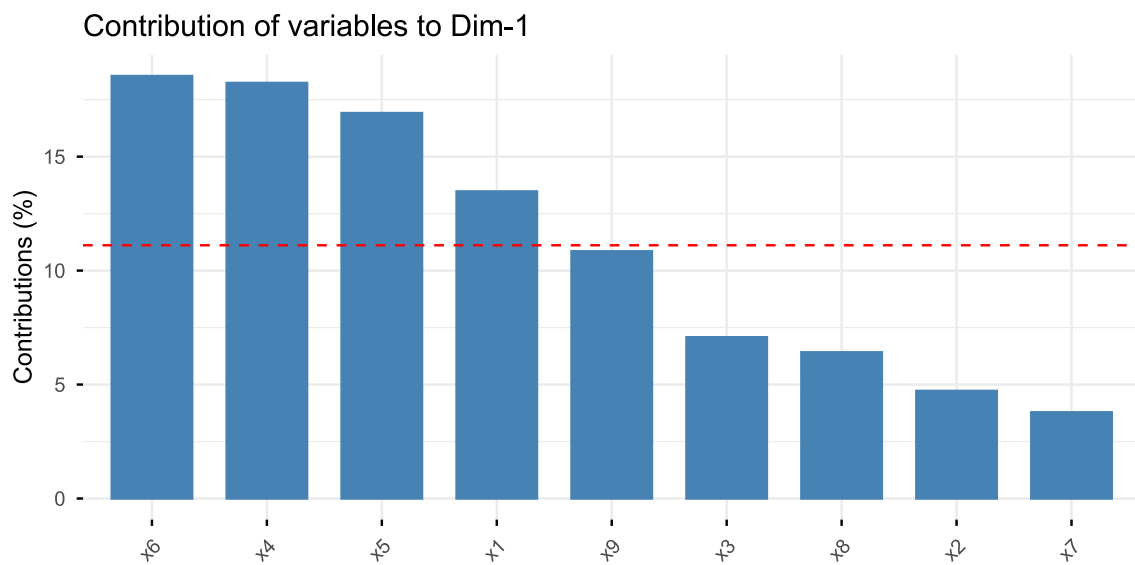
```
pca$var$cor[, 1:3] |> round(2)
```

```
      Dim.1 Dim.2 Dim.3
x1  0.66  0.13  0.37
x2  0.39  0.09  0.62
x3  0.48  0.33  0.54
x4  0.77 -0.45 -0.17
x5  0.74 -0.48 -0.22
x6  0.77 -0.43 -0.11
x7  0.35  0.50 -0.59
x8  0.45  0.61 -0.33
x9  0.59  0.51 -0.02
```

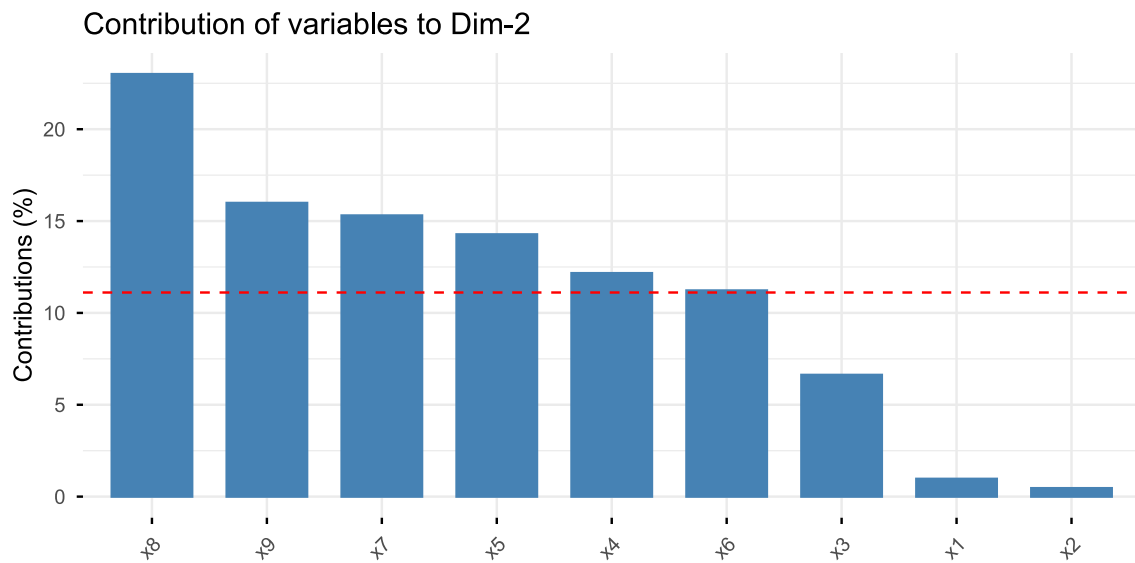
```
corrplot(pca$var$cos2[, 1:3], is.corr = FALSE)
```



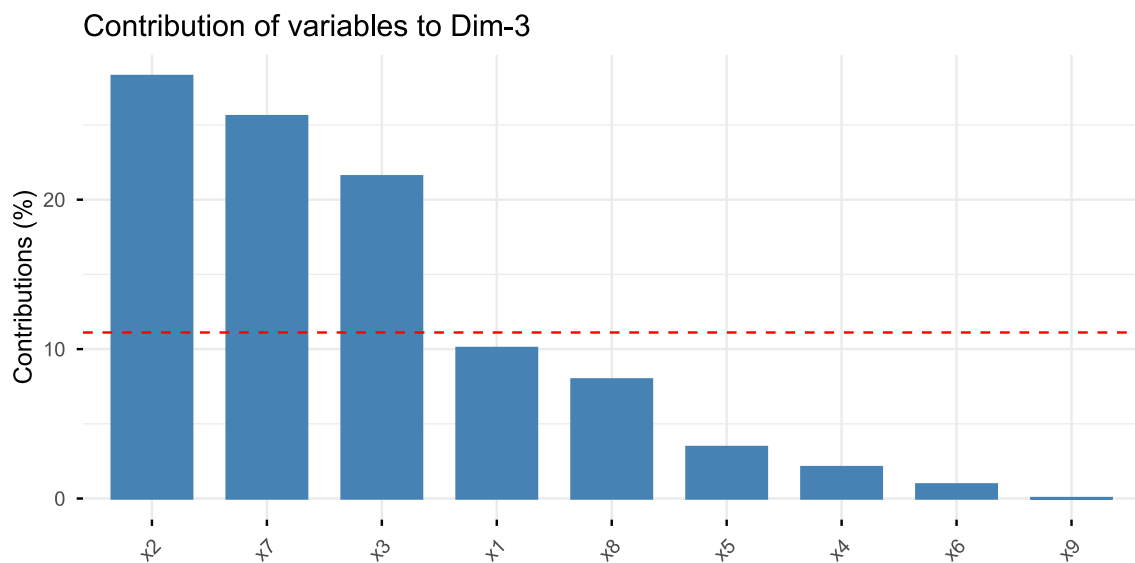
```
fviz_contrib(pca, choice = "var", axes = 1)
```



```
fviz_contrib(pca, choice = "var", axes = 2)
```



```
fviz_contrib(pca, choice = "var", axes = 3)
```

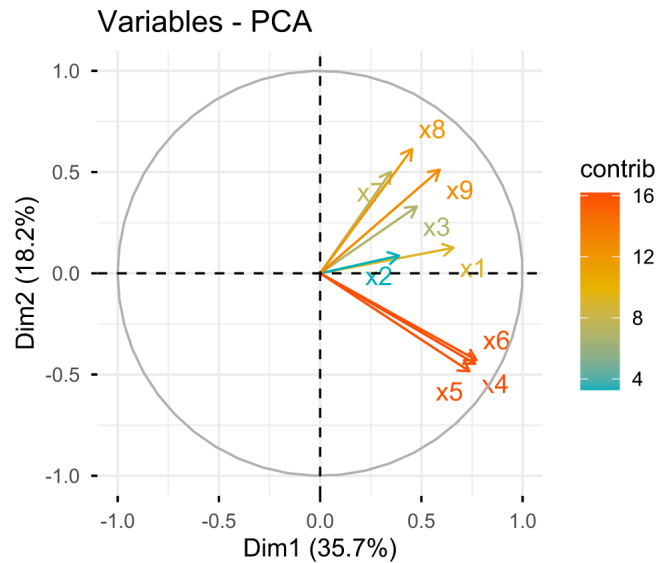


- **PC1 (35.7%), general ability:** all tests load positively (0.35 to 0.77), led by the verbal tests **x4**, **x5**, **x6**.
- **PC2 (18.2%), speed vs. verbal:** speed tests **x7**, **x8**, **x9** load positively (0.5 to 0.6), verbal tests negatively (about -0.45).
- **PC3 (15.2%), visual vs. speed:** visual tests **x1**, **x2**, **x3** load positively (0.37 to 0.62), **x7** and **x8** negatively.

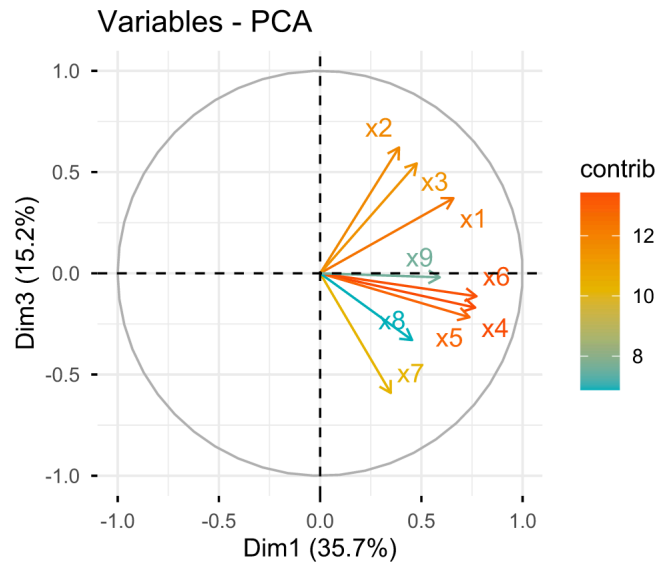
These visual, verbal and speed groups match the correlation blocks in the corplot.

2.6. Graphical representation

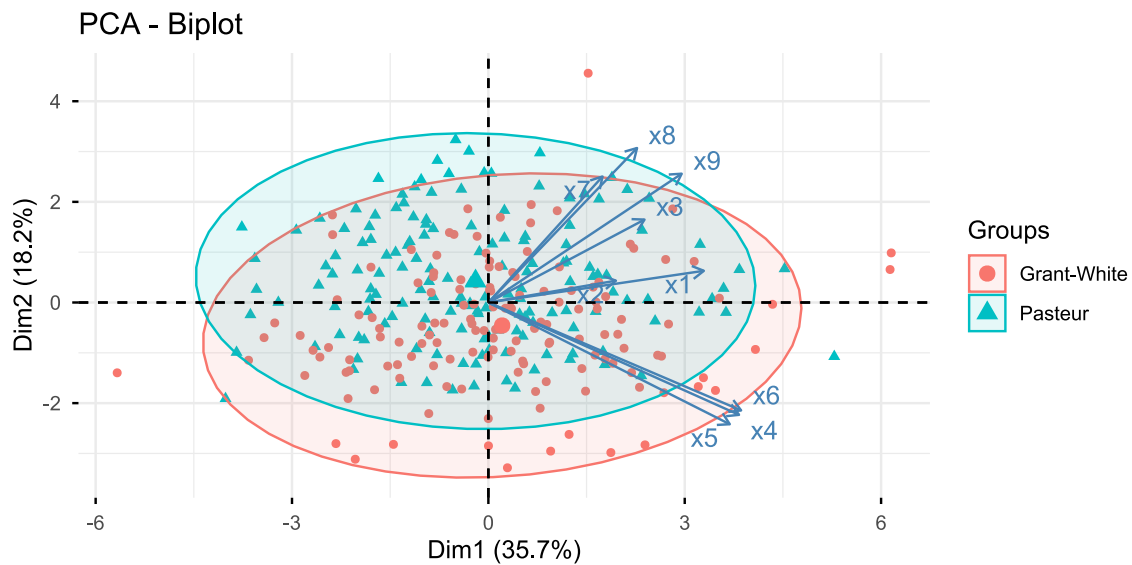
```
fviz_pca_var(pca, axes = c(1, 2), col.var = "contrib", repel = TRUE,
  gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"))
```



```
fviz_pca_var(pca, axes = c(1, 3), col.var = "contrib", repel = TRUE,
  gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"))
```



```
fviz_pca_biplot(pca, repel = TRUE, label = "var", col.ind = "grey70",
  habillage = data$school, addEllipses = TRUE)
```



The correlation circles show three variable groups (verbal, speed, visual). The biplot colours students by school to compare their scores.

3. Selecting for the Exploratory Factor Analysis (d)

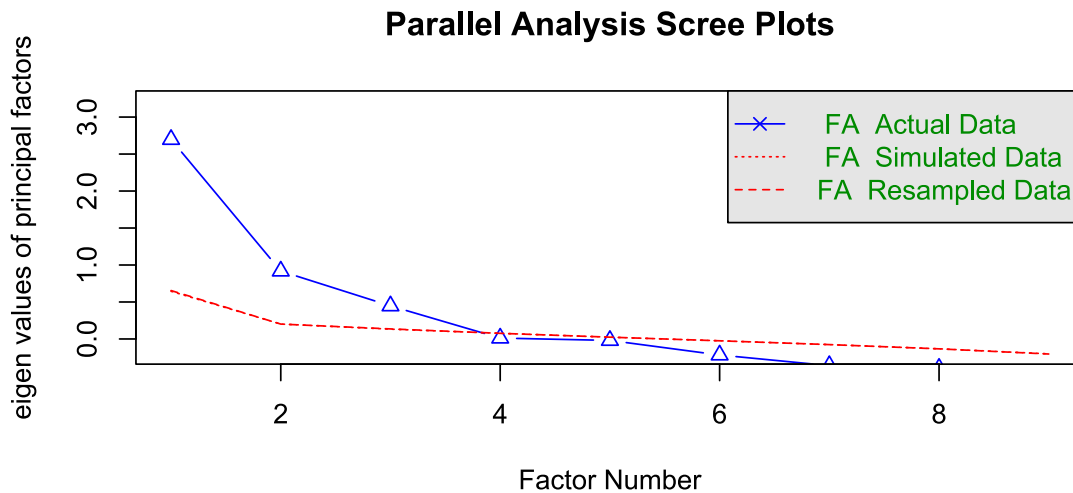
3.1. Number of factors

The PCA criteria are a starting point, but each has limits:

- **Kaiser** is simple but tends to keep too many components when there are many variables.
- **60% cumulative variance** is an arbitrary threshold and ignores whether a component is interpretable.
- **Horn's parallel analysis** compares against random data, so it is the most reliable of the three.

PCA eigenvalues use the total variance, while EFA models only the shared (common) variance. So we repeat parallel analysis on the factor model itself:

```
set.seed(123)
fa.parallel(df, fm = "ml", fa = "fa", n.iter = 500)
```



Parallel analysis suggests that the number of factors = 3 and the number of components = NA

It also suggests **3 factors**. With every criterion agreeing, and the correlation matrix showing three blocks, we extract 3 factors.

3.2. Which variables to keep

A variable is worth keeping if it shares variance with the others. We check:

- **Individual KMO (MSA):** should be above 0.6.
- **Communality:** how much of the variable the factors explain. Below about 0.2 is weak.
- **Loadings:** a variable should load clearly (above 0.4) on one factor.

```
KMO(df)$MSAi |> round(2)
```

```

x1  x2  x3  x4  x5  x6  x7  x8  x9
0.81 0.78 0.73 0.76 0.74 0.81 0.59 0.68 0.79

```

All MSA values are above 0.6 except **x7** (0.59), which is borderline. We keep it because it measures speed, one of the three abilities we expect, and later loads strongly on its factor. We therefore keep all nine variables.

4. Exploratory Factor Analysis (e)

4.1. Estimation

- **Extraction:** maximum likelihood (`fm = "ml"`), which also gives fit indices.
- **Rotation:** oblimin (oblique), because cognitive abilities are expected to correlate.

```
efa <- fa(df, nfactors = 3, fm = "ml", rotate = "oblimin")

print(efa$loadings, cutoff = 0.3, digits = 2)
```

```
Loadings:
      ML1  ML2  ML3
x1      0.60
x2      0.51
x3      0.69
x4  0.84
x5  0.89
x6  0.81
x7      0.72
x8      0.70
x9      0.37  0.46

      ML1  ML2  ML3
SS loadings  2.19 1.27 1.24
Proportion Var 0.24 0.14 0.14
Cumulative Var 0.24 0.39 0.52
```

```
efa$communality |> round(2)
```

```
  x1  x2  x3  x4  x5  x6  x7  x8  x9
0.49 0.25 0.46 0.72 0.76 0.69 0.50 0.53 0.46
```

```
efa$Phi |> round(2)
```

```
      ML1  ML2  ML3
ML1 1.00 0.33 0.22
ML2 0.33 1.00 0.27
ML3 0.22 0.27 1.00
```

4.2. Model fit

```
efa$TLI
```

```
[1] 0.963885
```

```
efa$RMSEA
```

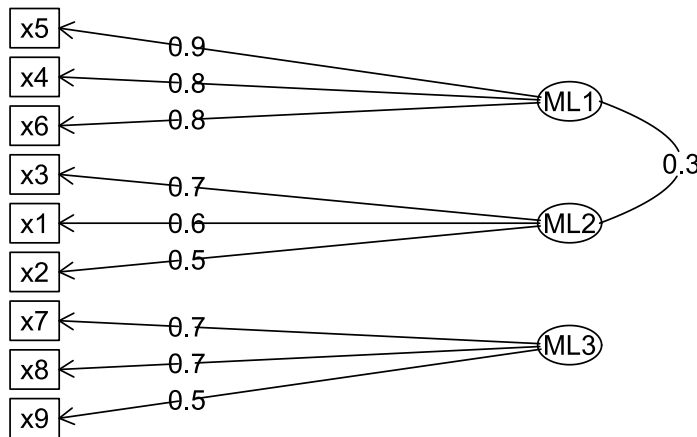
```
      RMSEA      lower      upper confidence  
0.05349608 0.01474829 0.08779583 0.90000000
```

TLI = 0.96 (good above 0.95) and RMSEA = 0.053 (acceptable below 0.06), so the three-factor model fits well. The chi-square test is significant ($\chi^2 = 22.4$, $df = 12$, $p = 0.034$), but it is sensitive to sample size. The three factors explain 53.9% of the total variance.

4.3. Interpreting the factors

```
fa.diagram(efa)
```

Factor Analysis



Factor	Variables (loading)	Interpretation
ML1	x4 (0.84), x5 (0.89), x6 (0.81)	Verbal ability
ML2	x1 (0.60), x2 (0.51), x3 (0.69)	Visual-spatial ability
ML3	x7 (0.72), x8 (0.70), x9 (0.46)	Processing speed

- **Verbal** is the strongest factor: paragraph comprehension, sentence completion and word meaning all load above 0.8.
- **Visual-spatial** groups the visual perception, cubes and lozenges tests.
- **Speed** groups the timed addition, dot counting and letter discrimination tasks.
- x9 cross-loads on the visual factor (0.37), since discriminating straight and curved letters needs both speed and visual perception.
- x2 has the lowest communality (0.25), so it is the weakest indicator.

- The factors correlate moderately (0.22 to 0.33), which supports the oblique rotation.

Compared with the PCA, the rotation gives a cleaner structure: each variable loads on one factor, instead of the contrasts seen in PC2 and PC3.